Name: **Dudipala Kavya Reddy**
Affiliation: Multi-disciplinary Natural Science Researcher
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Trainee Patent Attorney:

Company: Haley Guiliano LLP
Location: London, United Kingdom
Position: Technical Advisor
Work time: Part-time (20 hours per week)
Period: September 2023 – May 2024
Role: Patent prosecution under supervision of qualified UK and European patent attorneys. Worked on USA and European patents in drafting claims and drafting response letters to office actions for clients in semiconductor and media industries.

Education:

University: University of Oxford
Location: Oxford, United Kingdom
Department: Inorganic Chemistry
Group: Energy Materials and Devices group
Position: Materials Researcher (DPhil Inorganic Chemistry)
Period: Jan 2023 – Now (Feb 2025)
Role: Develop novel materials for ionizing radiation detection with focus on scaling up production and commercialization. Involves single crystal synthesis, semiconducting wafer fabrication, device fabrication, testing and optimization.

University: Imperial College London
Location: London, United Kingdom
Department: Materials Science and Engineering
Group: Energy Materials and Devices group
Position: Materials Researcher (PhD Materials)
Period: Nov 2021 – Jan 2023
Role: Develop advanced, high-performance semiconductors for energy applications. Involves materials synthesis, post treatment, material characterization, improving material properties, depositing thin films using solution processing, various PVD and CVD deposition techniques, device structure optimization, device fabrication in clean room environment, device characterization.

Materials synthesis and Characterisation:

- X-ray Diffraction
- Energy Dispersive X-ray spectroscopy
- Scanning Electron Microscope
- Transmission Electron Microscope
- UV-vis absorption spectroscopy
- Stationary and time-resolved florescence spectroscopy
- EPR Spectroscopy
- X-ray Photoelectron Spectroscopy
- Raman Spectroscopy
- Hall measurements & device performance
- Surface Profilometry
- Vickers micro-hardness & nano-indentation
- Scratch, wear, erosion tests
- Cathodic Arc Physical Vapor Deposition
- DC & RF sputtering
- Vacuum thermal evaporation
- E-beam evaporation
- Chemical Vapor Transport
- Spin coating & dip coating

University: Ludwig Maximillian University
Location: Munich, Germany
Department: Physics
Group: Photonics and Optoelectronics Group
Position: Erasmus+ Exchange student
Period: Oct 2019 – Aug 2020
Grade: Outstanding – 1.0 German Grade
Role: Perovskite semiconductors research for applications in opto-electronic devices. Involved semiconductor nanocrystal synthesis and studying their structural, morphological, and optical properties like absorption spectra, photoluminescence spectra, and quantum yield. Exciton behavior was studied to understand the trap states generated during material degradation. These photo-active nanocrystals were also investigated for their potential as photocatalysts.

University: Politecnico di Torino
Location: Turin, Italy
Department: Applied Science and Technology
Position: Voluntary visiting student
Period: Nov 2018 – May 2019
Role: Magnetron sputtered PVD thin films deposition and characterization for their application as decorative coatings for automobiles and anti-microbial coatings for hospital gowns and other health care textiles. Developing an alternative green solution to the current-day commercial chrome coatings. Developing a self-cleaning and anti-bacterial protective coating for textiles used in health care sector.

University: University of Turin
Location: Turin, Italy
Department: Chemistry
Degree: Master of Materials Science
Period: Oct 2018 – April 2021
Grade: Distinction (Honors) – 110/110 con Lode – Italian Grade
Course work: Quantum mechanics, Solid state physics (Semiconductors and Superconductors), Metallurgy, Polymeric materials with laboratory, Advanced Mathematics and numerical analysis, Advanced crystallography, Solid state chemistry with laboratory, Analytical chemistry for materials science, Physical chemistry, Selection and use of materials, master thesis on all-inorganic perovskite quantum dots to enhance their ambient stability.

Coursework labs:

- Semiconductors
- AC and Electronic circuits
- Electricity and Magnetism
- Qualitative analysis
- Quantitative analysis
- Optics and laser physics
- Quantum Mechanics
- Nuclear physics
- Physical Chemistry
- Analytical Chemistry
- Heat and Modern physics

(and more)

Programming languages:

- Python
- Matlab

Softwares:

- Microsoft suite
- Origin & Expert Highscore Plus
- Crystal Maker and Vesta
- Labview
- Solidworks

Languages:

- English: C1
- German, Italian: A1

Soft skills:

- Self-motivated
- Interpersonal Communication
- Team-work and collaboration
- Creativity and problem solving

University: University of Hyderabad (UoH)
Location: Hyderabad, India
Department: Physics
Degree: Master of Physics (5 year-Integrated)
Period: Aug 2012 – July 2017
Grade: First division – 7.78/10 CGPA
Course work: Mathematical Physics, Thermodynamics and statistical physics, Mechanics, Foundation Biology, Energetics and Kinetics, Structural Chemistry, Information Technology, Molecules and Information processing, Structure and function of Macro-molecules, Probability and Statistics, Classical mechanics, Quantum physics, Heat and Modern Physics, Nuclear physics, Atomic physics, Identification of organic compounds, Electromagnetic theory, Digital Electronics, Magnetism, Probes of Condensed matter physics, Statistical Mechanics, Laser physics, Particle physics, General properties of matter, Computational methods, Signals and systems, Environmental studies, Electronic devices and circuits, and more.

Workshop: European Innovation Academy
Type: Entrepreneurship program
Location: Turin, Italy
Period: 7^{th}-26^{th} July 2019
Contents: Team formation, Problem solving with design, New technologies, Problem-solution fit, Online validation and google tools, Stress management and communication in team work, unit economics, Product-market fit and market selection, customer persona and customer validation, prototyping, solution validation, design hacks, Neuromarketing, social network marketing, user acquisition, Growth hacking, product sprint, Launch tactics, Business and revenue model design and growth, fundraising, venture capital overview, patents and trademarks, meeting with investors, strategy for future.

R&D Centre: International Advanced Research Centre for Powder Metallurgy and New Materials (ARCI)
Centre: Centre for Engineered Coatings
Position: Research Intern
Period: May 2016 – July 2016
Role: Design and develop wear-resistant, adhesive coatings for die casting applications, using Cathodic Arc PVD. Perform several thin film characterization techniques to understand physical properties of the deposited films.

Transferable skills

- Technical expertise: in-depth knowledge in specific fields of science and engineering
- Analytical and critical thinking: evaluating complex problems and problem solving
- Research skills: scientific acumen gained through a decade of experience
- Writing and communication: comprehension and articulation of complex technical concepts to variety of audiences in clear and precise manner
- Attention to detail: precision and meticulousness to ensure accuracy and avoid errors
- Time management: managing multiple projects and meeting deadlines simultaneously
- Strategic thinking and adaptability: ability to navigate through unforeseen challenges and adapt to changing circumstances
- Presentation and public speaking skills: ability to explain concepts and defend views with experts in field
- Collaboration and team work: to leverage diverse expertise and perspectives
- Continuous learning: ability to sustain intellectual curiosity and staying updated with industry advancements
- Client management, mediation and negotiation: collaborative research with negotiation and conflict resolution skills
- Resilience: patience, determination and resilience to overcome setbacks